

Suppose a sphere $\mathcal{S}$ of radius 1 centred at point P with position vector

$$\underline{p} = \begin{bmatrix} 10 \\ 0 \\ 2 \end{bmatrix},$$

and define the line

$$\ell = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} + k \begin{bmatrix} 1 \\ Y \\ Z \end{bmatrix},$$

where $k \in \mathbb{R}$ and $Y, Z \sim \mathcal{U}[-1, 1]$.

- (i) By considering an appropriate perpendicular-distance expression, deduce that for ℓ to intersect $\mathcal{S}$,

$$\frac{10\sqrt{Y^2 + Z^2}}{\sqrt{1 + Y^2 + Z^2}} \leq 1,$$

and further show that

$$Y^2 + Z^2 \leq \frac{1}{99}.$$

- (ii) Explain why this result is meaningful. **You may assume the fact that for $X, Y \sim \mathcal{U}[-n, n], n \in \mathbb{R}^+$, the point $P(X, Y)$ is uniformly distributed in the resulting square without proof.**
- (iii) Hence, by considering the ratio of favourable outcomes to total outcomes, deduce the probability that the line ℓ intersects $\mathcal{S}$.